



```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
In [ ]:
```

```
In [13]: x=np.array([95,85,80,70,60])
y=np.array([85,95,70,65,70])
```

```
In [14]: model= np.polyfit(x, y, 1)
```

```
In [15]: model
```

```
Out[15]: array([ 0.64383562, 26.78082192])
```

```
In [16]: predict = np.poly1d(model)
predict(65)
```

```
Out[16]: 68.63013698630137
```

```
In [17]: y_pred = predict(x)
y_pred
```

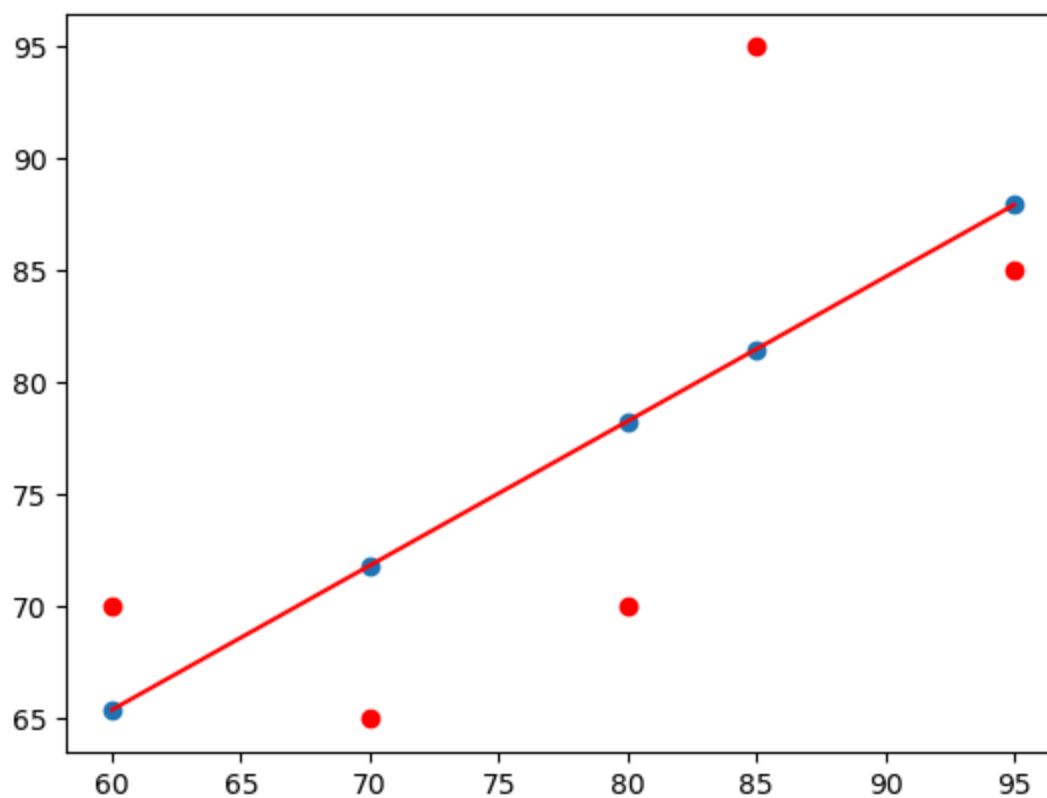
```
Out[17]: array([87.94520548, 81.50684932, 78.28767123, 71.84931507, 65.4109589 ])
```

```
In [18]: from sklearn.metrics import r2_score
r2_score(y, y_pred)
```

```
Out[18]: 0.4803218090889326
```

```
In [19]: y_line = model[1] + model[0]* x
plt.plot(x, y_line, c = 'r')
plt.scatter(x,y_pred)
plt.scatter(x,y,c='r')
```

```
Out[19]: <matplotlib.collections.PathCollection at 0x1bf12a0f110>
```



```
In [20]: import numpy as np  
import pandas as pd  
import matplotlib.pyplot as plt
```

```
In [2]: df=pd.read_csv("BostonHousing.csv")
```

```
In [3]: df
```

Out[3]:

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	
0	0.00632	18.0	2.31	0	0.538	6.575	65.2	4.0900	1	296	15.3	396
1	0.02731	0.0	7.07	0	0.469	6.421	78.9	4.9671	2	242	17.8	396
2	0.02729	0.0	7.07	0	0.469	7.185	61.1	4.9671	2	242	17.8	396
3	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	222	18.7	396
4	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	222	18.7	396
...	...	...	...	...	...	...	...	...	...	...	...	...
501	0.06263	0.0	11.93	0	0.573	6.593	69.1	2.4786	1	273	21.0	396
502	0.04527	0.0	11.93	0	0.573	6.120	76.7	2.2875	1	273	21.0	396
503	0.06076	0.0	11.93	0	0.573	6.976	91.0	2.1675	1	273	21.0	396
504	0.10959	0.0	11.93	0	0.573	6.794	89.3	2.3889	1	273	21.0	396
505	0.04741	0.0	11.93	0	0.573	6.030	80.8	2.5050	1	273	21.0	396

506 rows × 14 columns

```
In [6]: x = df.drop(['medv'], axis = 1)
        y = df['medv']
```

```
In [7]: from sklearn.model_selection import train_test_split
        xtrain, xtest, ytrain, ytest = train_test_split(x, y, test_size
        =0.2, random_state = 0)
```

```
In [8]: from sklearn.linear_model import LinearRegression
        lm = LinearRegression()
        model = lm.fit(xtrain, ytrain)
```

```
In [9]: ytrain_pred = lm.predict(xtrain)
        ytest_pred = lm.predict(xtest)
```

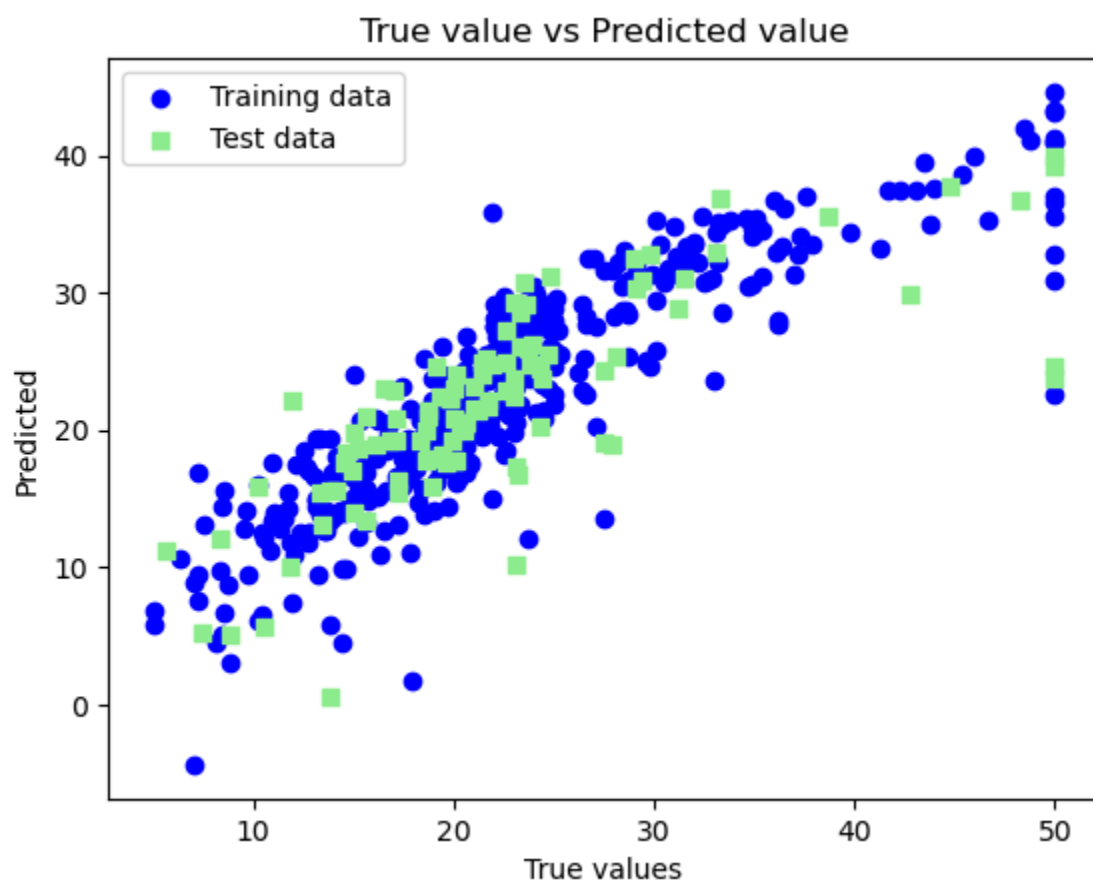
```
In [10]: df1 = pd.DataFrame(ytrain_pred, ytrain)
        df1 = pd.DataFrame(ytest_pred, ytest)
```

```
In [11]: from sklearn.metrics import mean_squared_error, r2_score
        mse = mean_squared_error(ytest, ytest_pred)
        print(mse)
        mse = mean_squared_error(ytrain_pred, ytrain)
        print(mse)
```

```
33.44897999767632
19.326470203585725
```

```
In [12]: plt.scatter(ytrain , ytrain_pred, c='blue', marker='o', label='Training data')
        plt.scatter(ytest, ytest_pred, c='lightgreen', marker='s', label='Test data')
        plt.xlabel('True values')
```

```
plt.ylabel('Predicted')
plt.title("True value vs Predicted value")
plt.legend(loc='upper left')
plt.plot()
plt.show()
```



In []: